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Enhanced coke resistant Ni/SiO₂@SiO₂ core-shell nanostructured catalysts for dry reforming of methane: Effect of metal-support interaction and SiO₂ shell



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Highlights

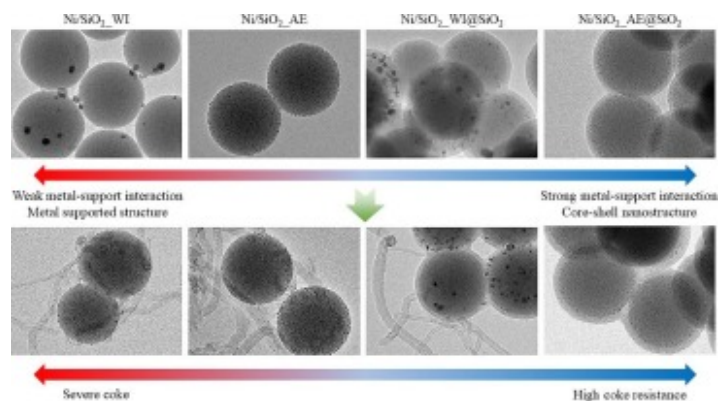
- Ni/SiO₂ catalysts were prepared by wet impregnation and ammonia evaporation methods.
- Ni phyllosilicate induced high dispersed Ni due to strong interaction with support.

- Ni/SiO₂@SiO₂ core shell structure was successfully prepared by sol-gel method.
- The highly dispersed Ni and confining effect cause excellent coke resistance.

Abstract

Suppressing coke formation on nickel-based catalysts used in methane-dry reforming is one of the most important issues. We investigate the effect of nickel metal-support interaction and outer porous silica shells on the coke formation in methane dry reforming over Ni/SiO₂@SiO₂ core-shell catalysts. Using a simple ammonia evaporation method, the Ni/SiO₂_AE catalyst, which has highly dispersed nickel nanoparticles with ca. 7 nm, was synthesized and coated with an outer porous silica shell to produce a Ni/SiO₂_AE/SiO₂ core shell catalyst. Based on the TEM and N₂ isotherm results, a 20 nm thick SiO₂ shell having pores with a size of about 2 nm was uniformly coated on the Ni/SiO₂_AE catalyst. H₂-TPR, and XPS analysis results suggest that the Ni/SiO₂_AE and Ni/SiO₂_AE@SiO₂ catalysts have high nickel dispersion. This high dispersion is due to strong metal-support interactions and the formation of Ni phyllosilicate species. On the other hand, the ones synthesized by wet impregnation method have low nickel dispersion due to weaker metal-support interactions. The Ni/SiO₂_AE@SiO₂ core-shell catalyst showed stable catalytic activity under methane dry reforming conditions at 600 °C. On the other hand, the Ni/SiO₂_WI and Ni/SiO₂_AE catalysts showed rapid deactivation due to severe coke formation. It can be concluded that the strong metal-support interaction and the silica shell can suppress catalyst deactivation caused by coke formation very effectively.

Graphical abstract



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Keywords

Dry reforming; Carbon dioxide; Core-Shell; Carbon deposition; Ni/SiO₂ catalyst

1. Introduction

After the industrial revolution in the 19th century, carbon dioxide emissions have increased as fossil fuel usage has increased exponentially (Friedlingstein et al., 2022). Carbon dioxide emitted from fossil fuels usage is well known as a main cause of global warming and climate change. A lot of research and technical development on renewable energy, such as solar power and wind power generations, have been carried out replace fossil fuels, but fossil fuels are still the main energy resource. Even though there have been a lot of progress in reducing carbon dioxide emission during the last 30years, we are still emitting a huge

amount of carbon dioxide. Since we cannot dramatically change our energy consumption pattern in modern society, in the current situation, it is essential to capture, utilize and store of carbon dioxide.

Recently, many researchers are paying attention on how to utilize the captured carbon dioxide. Among the methods that can utilize carbon dioxide, the chemical conversion of carbon dioxide into useful chemical substances is an attractive method ([Liu et al., 2023](#), [Yao et al., 2023](#)). Dry reforming of methane (DRM) (Eq. (1)) has been widely studied as a reaction for producing syngas by reacting methane with carbon dioxide, which are representative greenhouse gases. Syngas produced by DRM can be further converted into high-value chemical substances such as hydrogen, ammonia, methanol, and liquid fuel through water gas shift, ammonia synthesis, methanol synthesis, Fisher-Tropsch reaction, etc.

It is well known that precious metal-based catalysts such as the ones based on Ru ([Chen et al., 2010](#)), Rh ([Drif et al., 2015](#)), Pd ([Singha et al., 2017](#)), and Pt ([Damyanova et al., 2009](#)) can have high catalytic activity as well as exceptional durability against catalyst deactivation in a DRM reaction. Although a precious metal-based catalyst shows high catalytic activity, transition metal-based catalysts are being widely studied for the DRM reaction since the economic feasibility of transition metal is significant in a practical reactor ([San-José-Alonso et al., 2009](#), [Theofanidis et al., 2015](#), [Tsoukalou et al., 2016](#)). In particular, nickel-based catalysts are widely researched because they have high catalytic activity and lower cost compared to precious metal-based catalysts. However, nickel-based catalysts are known to suffer significant deactivation during DRM reaction due to carbon deposition through the methane decomposition reaction (Eq. (2)) and the boudouard reaction (Eq. (3)) ([Foppa et al., 2016](#), [Kawi et al., 2015](#)). The carbon deposition on the catalyst causes clogging of the reactor and makes the process operation difficult ([Ochoa et al., 2020](#)). In addition, the thermal sintering phenomenon of nickel-based catalysts during DRM reaction at high temperatures is also a factor that gradually deactivates the catalysts. Thermal sintering of the catalyst not only reduces the dispersity of the supported metal particles, but also causes additional problems such as carbon deposition. Carbon deposition reactions, especially the methane decomposition reaction, have been reported to be easily occurring on the surface of relatively large nickel particles (>7 nm) ([Kim et al., 2000](#)). Therefore, it is very important to disperse nickel nanoparticles efficiently in a methane dry reforming catalyst.



Since coke formation on small nickel particles is an energetically unfavorable reaction, much effort has been made to adjust the size of the nickel particles to improve the coking durability of the nickel-based catalysts in DRM reaction ([Kim et al., 2000](#)). Two main strategies have been used to adjust the size of metal particles. The first is a method of modifying the surface of the support. Xiangyu et al. ([Lv et al., 2012](#)) used ethylene glycol treatment to the silica support to remove H-bond SiOH, which causes the formation of large metal particles. In an ethylene glycol-treated Ni/SiO₂ catalyst, the metal support interaction improved, and as a result, nickel was highly dispersed. Also, since a negative charge is formed on the surface of SiO₂ under basic conditions, a highly dispersed catalyst can be produced by strong electrostatic adsorption (SEA) of a nickel precursor on the SiO₂ support under basic conditions. Fagen et al. ([Wang et al., 2021](#)) successfully prepared a Ni/SiO₂ catalyst by SEA method (Ni/SiO₂-E) with smaller size

of nickel than that of the catalyst produced by wet impregnation method (Ni/SiO₂-I). As a result, the Ni/SiO₂-E catalyst showed high activity, catalytic stability, and coke durability in the low temperature DRM reaction. The second method is to incorporate nickel particles into the support. A nickel-incorporated catalyst can be produced by introducing a nickel precursor during the oxide support (SBA-15, Al₂O₃) synthesis process (Cai et al., 2014, He et al., 2021, Wang et al., 2017). Once Ni was incorporated into the support, strong metal support interactions allowed the small particle size to be maintained at high temperatures, reducing coke formation. Wenjia et al. (Cai et al., 2014) claimed that these strong metal-support interactions were evoked by the anchoring effect between metallic nickel particles and unreduced nickel ions in the silica matrix. In conclusion, both methods controlled the dispersity of the metal particles through controlling the interaction between the metal and the support.

However, in the case of a general metal-supported catalyst, thermal sintering cannot be completely suppressed under high-temperature DRM reaction conditions (Zhao et al., 2018). Forming core-shell nanostructure in which the metal nanoparticles are encapsulated by a support shell is one of the most powerful strategies to prevent sintering of active metal particles at high temperatures through completely separating the metal nanoparticles (Joo et al., 2009). Once we introduce the porous support outer shell in the core-shell nanostructures, reactant molecules can selectively diffuse to the active metal surface, but metal particle cannot be moved resulting in preventing thermal metal sintering.

So far, core-shell nanostructured catalysts in which active metals are encapsulated in the porous out layer have been suggested and successfully applied to many important catalytic reactions. A lot of versions of Ni based core-shell catalysts are also investigated in high temperature catalytic reactions including steam reforming of hydrocarbon (SR) (Pu et al., 2018a, Pu et al., 2018b), DRM (Bian et al., 2022, Kaviani et al., 2022) and photothermal DRM (Han et al., 2021). Although core-shell catalysts using various support shells such as Ni@TiO₂ (Zhao et al., 2015), Ni@Al₂O₃ (Pu et al., 2018a, Pu et al., 2021, Yao et al., 2010), Ni@ZrO₂ (Lim et al., 2021), and Ni@CeO₂ (Tang et al., 2017) have been studied, Ni@SiO₂ core-shell catalysts including silica shells have been the most widely studied. This is because the silica shell is easy to be synthesized and to be finely controlled through sol-gel chemistry. Zhao et al. (Zhao et al., 2018) prepared a core-shell catalyst encapsulating silica shell over nickel and nickel-cobalt bimetallic nanoparticles and showed a stable catalytic activity without sintering and coking in a methane dry reforming reaction at 800°C. Shi et al. (Shi et al., 2022) and Han et al. (Han et al., 2022) reported nickel-copper alloy based core shell catalysts, (Ni-Cu/CeO₂)@SiO₂ and Ni-Cu@SiO₂, respectively. They reported that Cu promoted CO₂ activation and improved CH₄ activation by modifying the electronic environment of Ni in Ni-Cu alloy catalyst, resulting in high and stable catalytic performance in dry reforming reaction. Ziwei et al. (Li et al., 2014) prepared a Ni-yolk@Ni@SiO₂ catalyst by coating a silica shell with a thickness of 10 to 15 nm on nickel nanoparticles. However, the above processes for synthesizing small and uniformly sized nickel nanoparticles

are quite complicated and use toxic and expansive organic solvents and surfactants (Donegan et al., 2012, Hou et al., 2005, Park et al., 2005), which can induce safety issues in mass production. Therefore, it is necessary to investigate a facile synthesis method which can synthesize a core-shell catalyst having small nickel nanoparticles with uniform size by simple processes and non-toxic chemicals.

In this study, we synthesized the silica encapsulated Ni/SiO₂@SiO₂ core-shell nanostructured catalyst for enhancing the resistance to coke formation in DRM conditions. During the catalyst preparation, Ni/SiO₂ samples with nickel loaded on the surface of spherical silica were prepared using either wet impregnation or ammonia evaporation methods to control the size of nickel nanoparticles. Then, a porous silica layer was coated on Ni/SiO₂ to prepare the Ni/SiO₂@SiO₂ catalyst having a core-shell structure. We applied both Ni/SiO₂ catalyst and Ni/SiO₂@SiO₂ catalyst prepared by two different synthesis methods to dry reforming of methane and investigated the effects of metal-support interaction of nickel nanoparticles and outer silica shell on the catalytic performances, especially the resistance to coke formation.

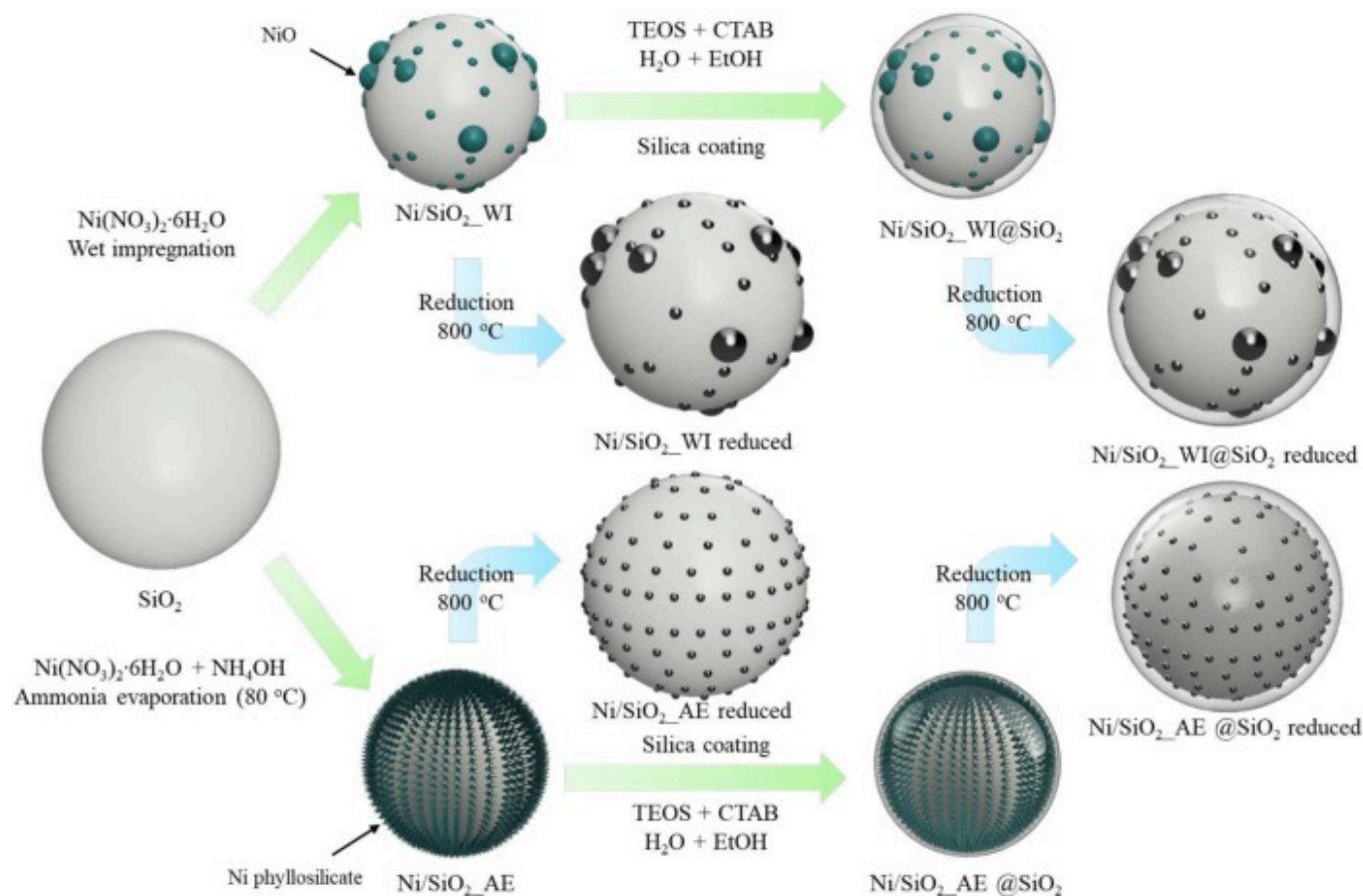
2. Experimental

Materials: Ethyl alcohol anhydrous (C₂H₆O, 99.9%), ammonium hydroxide aqueous solution (NH₄OH, 28%), nickel nitrate hexahydrate (Ni(NO₃)₂ 6H₂O, 97%) and cetyltrimethylammonium bromide (CTAB, C₁₉H₄₂BrN, 99%) were obtained from Daejung Chemical Company. Tetraethylorthosilicate (TEOS, SiC₈H₂₀O₄, 99%) was obtained from Alfa Aesar.

Preparation of silica sphere: Silica sphere was synthesized by the modified stobe method. To prepare silica sphere, ethanol (92ml), D.I water (16ml), NH₄OH (3ml) and TEOS (3ml) were mixed in a flask and stirred for 3h.

Preparation of Ni/SiO₂ catalysts: Ni/SiO₂ catalysts are synthesized by either a wet impregnation or an ammonia evaporation using pre-synthesized silica sphere. In addition, Ni/SiO₂@SiO₂ are prepared by coating of outer silica layer on the above Ni/SiO₂ catalysts through sol-gel chemistry. The overall synthesis route of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts is illustrated in [Scheme 1](#). The synthetic details are followings. To control the Ni dispersion of the catalysts, the Ni/SiO₂ catalysts was prepared by two different synthesis method such as wet impregnation and ammonia evaporation. For the wet impregnation method, prepared silica sphere was recovered by centrifugation, followed by impregnation with nickel nitrate aqueous solution. After impregnation process, resulting solid was dried and calcined at 700°C for 3h. For the ammonia evaporation method, nickel nitrate was added to the above silica dispersed solution. After a vigorous stirring for a couple of minutes, the solution was heated at 80°C for 3h to

evaporate the ammonia under stirring. After purple colored suspension transformed into a green suspension, green precipitate was recovered by centrifugation followed by washing with ethanol. After ammonia evaporation process, resulting solid was dried and calcined at 700 °C for 3 h. The Ni/SiO₂ catalysts prepared by wet impregnation method and ammonia evaporation method were denoted as Ni/SiO₂_WI and Ni/SiO₂_AE, respectively.



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Scheme 1. Synthesis route of Ni/SiO₂, Ni/SiO₂@SiO₂ catalysts.

Preparation of Ni/SiO₂@SiO₂ catalysts: To prepare Ni/SiO₂@SiO₂ catalysts, an outer silica layer is additionally coated on the above two Ni/SiO₂ catalysts by the sol-gel method. Each of the above two Ni/SiO₂ catalysts (0.2g), CTAB (0.4g), ethanol (80ml)

and D.I water (30ml) were mixed and sonicated until the Ni/SiO₂ catalysts were completely dispersed. After sonication, NH₄OH (2ml), the ethanolic solution of TEOS (0.4ml), and ethanol (40ml) were added to the above mixture and the final solution was stirred for 3h. The resulting solid was recovered by centrifugation, then washed by ethanol. After silica shell coating process, the solid precipitates were dried and calcined at 700°C for 3h. Ni/SiO₂@SiO₂ catalysts with Ni/SiO₂_WI or Ni/SiO₂_AE as core material were denoted as Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂, respectively.

Characterization: The pore structure characterization of sample was carried out by nitrogen adsorption isotherm at -196°C on Tristar II 3020 (Micromeritics). The pore size distribution and pore volume were obtained by BJH method from desorption branches of nitrogen isotherm and adsorbed volume at $P/P_0=0.99$. The X-ray diffraction spectrum was obtained on D/MAX 2200 (Rigaku) in the 2θ range of 20° to 80°. The crystalline size of Ni and NiO was calculated by Scherrer equation. The temperature programmed reduction (H₂-TPR) measurement was conducted on BELCAT-M (Microtrac BEL). For TPR measurement, 50mg of catalyst was pretreated in argon flow for 30min at 200°C. After cooled down to room temperature, the catalyst was programmatically heated to 1000°C at 5°C/min in 30ml/min of 5% H₂/Ar gas stream. The thermogravimetry analysis was accomplished using TGA N-1000 (Scinco) to weight the amount of coke in spent catalyst. The mass change was measured in an air atmospheric condition and temperature range of 30 to 800°C with a heating rate of 10°C/min. Transmission electron microscope (JEM-2100, JEOL) was used to analyze the morphology of catalysts. The amount of Ni loading was measured by using a Thermo Fisher iCAP 6300 inductively coupled plasma optical emission spectrometer (ICP-OES). The X-ray photoelectron spectra (XPS) were recorded by using X-ray photoelectron spectrometer (K-Alpha XPS system, Thermo Fisher Scientific). Nickel dispersion was determined by H₂ pulse chemisorption (BELCAT-M, Microtrac BEL). To conduct H₂ pulse chemisorption, all the catalyst were reduced at 800°C for 1 h under 4% H₂/Ar flow followed by cooling down to room temperature under Ar stream. Then H₂ pulses were injected by 6-port valve until the catalyst surface was saturated by hydrogen adsorption. The dispersion of Ni was calculated by assuming a stoichiometry factor of 2.

Catalytic activity test: Catalytic activity test was conducted in a fixed bed reactor at 101.3kPa. The 0.05g of catalysts were loaded in a quartz tube (ID=7 mm) between quartz wool plug. The gas volumetric flow rate was controlled by a mass flow controller. Before conducting the catalytic activity test, the catalysts were reduced under a stream of 33% of H₂/N₂ mixture gas with a flow rate of 30ml/min at 800°C for 1 h. The methane dry reforming reactions were carried out with feed gas flow rate of 40ml/min with compositions of N₂:CH₄:CO₂=2:1:1. The outflow gases were analyzed by on-line GC-TCD (YL-6500, Young Lin instrument) using Carboxen 1000 packed column. Ar carrier gas was used to analyze CH₄, CO₂, N₂, CO, H₂ in effluent gas. Concentration of CH₄, CO₂, H₂, CO in the product gas was calculated based on N₂, as an internal standard. The conversion rate of CH₄ and CO₂ (X_{CH_4}

and X_{CO_2}), H_2/CO ratio and turnover number of CH_4 and CO_2 are defined by the following equations:

$$X_i (\%) = \frac{F_{i_{in}} - F_{i_{out}}}{F_{i_{in}}} \times 100 \quad (4)$$

$$H_2/CO = \frac{F_{H_2_{out}}}{F_{CO_{out}}} \quad (5)$$

$$Specific\ Activity = \frac{F_{i_{in}} \cdot X_i}{w_{cat} \cdot C_{Ni}} \quad (6)$$

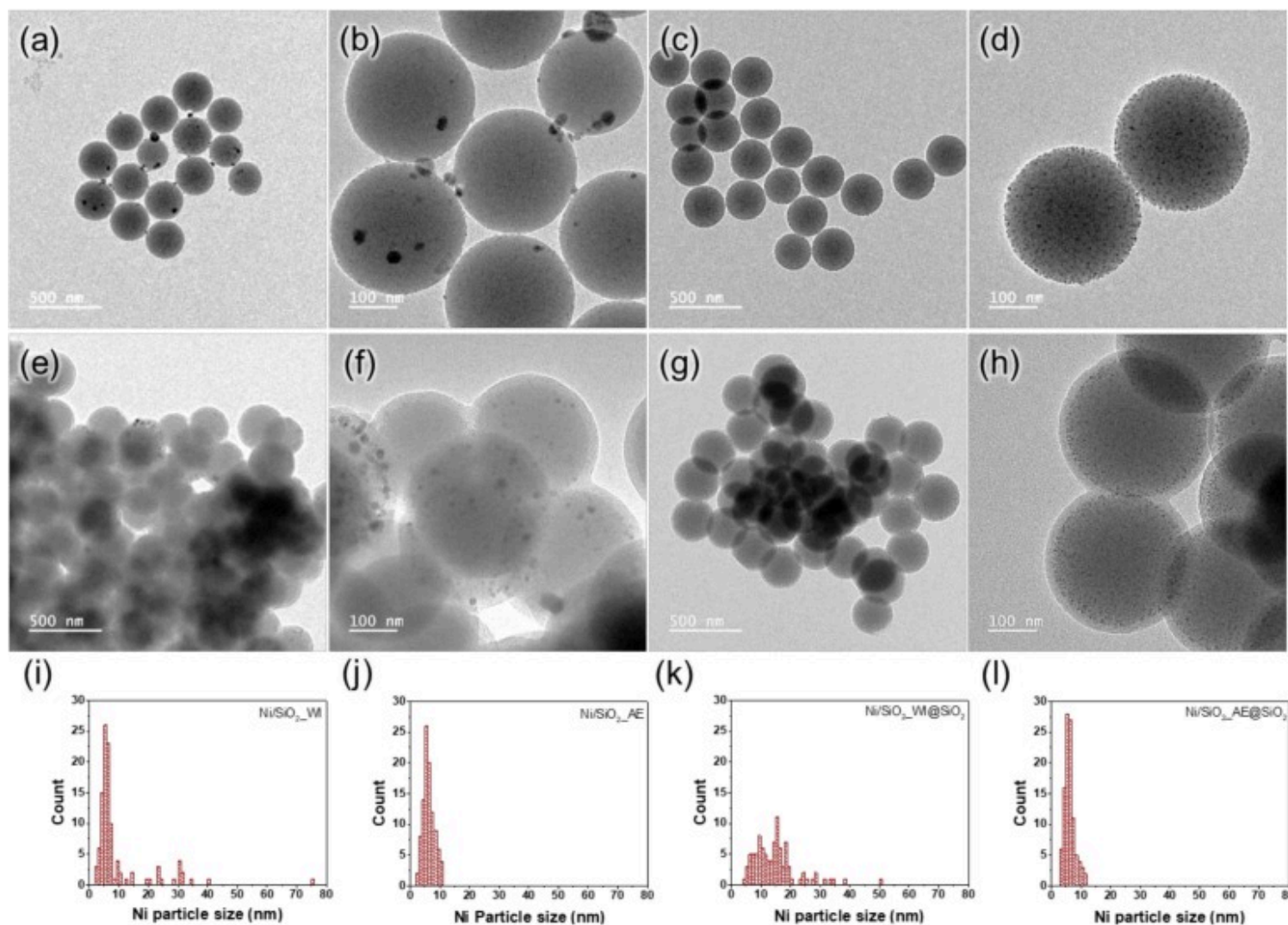
where X_i is the conversion of CH_4 and CO_2 , W_{cat} is the weight of the catalyst, C_{Ni} is the weight percent of loaded nickel, and F is the molar fraction.

3. Results and discussion

Characterization of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts: The morphologies of the calcined and reduced catalysts were characterized by TEM. As shown in Fig. S1(a–d), the TEM images of the calcined Ni/SiO₂_WI and Ni/SiO₂_AE catalysts showed the uniform size of core SiO₂ particles well prepared by the stober method. It can be easily observed that the thermally sintered NiO particles with un-uniform size are dispersed on SiO₂ nanospheres in the calcined Ni/SiO₂_WI catalyst (Fig. S1(b)). However, Ni/SiO₂_AE showed needle like branches on the surface of the SiO₂ sphere, known as nickel phyllosilicate (Fig. S2) (Ashok et al., 2018). After an additional SiO₂ coating, both the calcined Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂ exhibited an outer shell with less contrast, indicating that the porous silica shell layers are successfully coated on the surface of the Ni/SiO₂ catalysts (Fig. S1(e–h)).

Fig. 1 shows the TEM images of the reduced Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts. The TEM analysis was carried out after reduction at 800 °C. As shown in Fig. 1(a–d), the reduced nickel nanoparticles were well deposited on the surface of silica spheres. The Ni/SiO₂_WI catalysts have nickel nanoparticles of non-uniform size ranging from 5 to 65 nm (Fig. 1(a and b)). However, the Ni/SiO₂_AE catalysts have nickel nanoparticles of size 5–10 nm, smaller than those in the Ni/SiO₂_WI catalyst, dispersed uniformly on the surface of the silica sphere (Fig. 1(c and d)). In addition, the needle-like branches of nickel phyllosilicate species were converted into small Ni nanoparticles on the surface on SiO₂ by H₂ reduction (Fig. 1(d)). Both reduced Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂ showed that the porous silica layer is uniformly formed with approximately 20 nm thickness (Fig. 1(e–h)). It means that the silica shell layers are well maintained during the sol–gel SiO₂ coating followed by high temperature calcination

and reduction. In addition, it can be easily observed that all the nickel nanoparticles are encapsulated by the porous silica layer in both reduced Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂. Similar to the above calcined samples, the reduced Ni/SiO₂_WI@SiO₂ showed relatively large Ni particles between the inner silica core and the outer silica shell while the reduced Ni/SiO₂_AE@SiO₂ exhibited small sized Ni particles with uniform dispersion. Since the outer silica layer can play a role as a porous physical barrier, even though the reactant molecules can freely access the Ni surface through the mesopore, it can be expected that the thermal sintering of Ni nanoparticles are efficiently avoided under even high temperature DRM conditions.

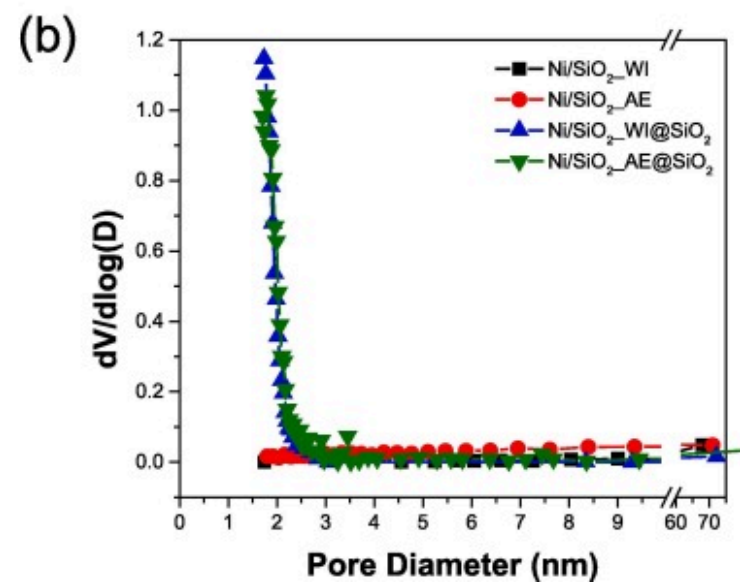
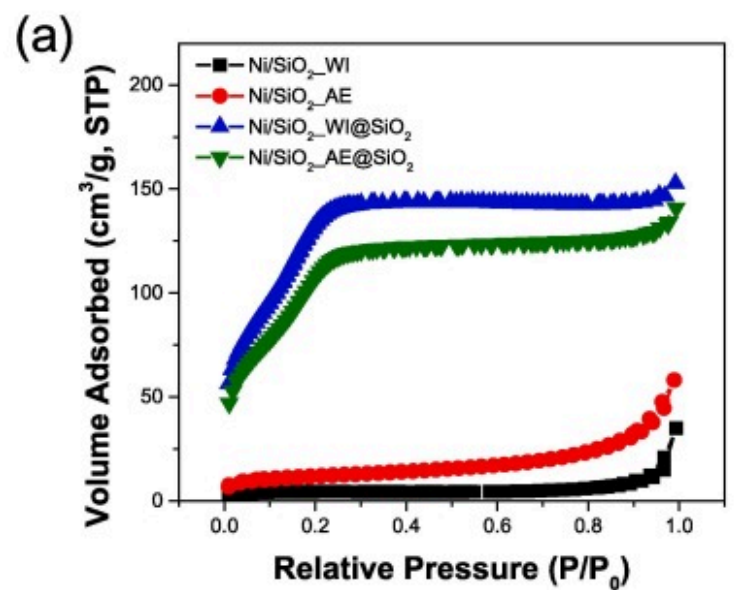


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Fig. 1. TEM image of (a, b) Ni/SiO₂_WI; (c, d) Ni/SiO₂_AE; (e, f) Ni/SiO₂_WI@SiO₂; (g, h) Ni/SiO₂_AE@SiO₂ after reduction at 800 °C and Ni particle size distribution of (i) Ni/SiO₂_WI, (j) Ni/SiO₂_AE, (k) Ni/SiO₂_WI@SiO₂ and (l) Ni/SiO₂_AE@SiO₂ measured by TEM.

Fig. 2 shows the results of N₂ physisorption and pore size distribution of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts. The Ni/SiO₂ catalysts have a type II isotherm and no obvious hysteresis loop. It means that the Ni/SiO₂ catalysts have non-porous characteristics. The specific surface area of Ni/SiO₂_AE is 40.4 m²/g, which is slightly larger than that of Ni/SiO₂_WI (13.4 m²/g). During Ni deposition through ammonia evaporation, some Ni species can be reacted with silica to form nickel phyllosilicate and it allows the surface of silica chemically etched out. On the other hand, Ni precursor in the wet impregnation is purely converted to NiO by calcination. In addition, the Ni/SiO₂_AE has slightly higher surface area than the Ni/SiO₂_WI. Additionally, the pore size of Ni/SiO₂_AE is 11.8 nm, which is consistent with the pore size derived from the needle-like branch of nickel phyllosilicate, as shown in previous studies (Table 1) (Zhang et al., 2013). The Ni/SiO₂@SiO₂ catalysts show type I isotherms and continuous adsorption until the relative pressure is below 0.25 (Fig. 2(a)). It indicates that both Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂ catalysts have mesopores with a size of ca. 2 nm. This can be seen in the pore size distribution. Both Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂ catalysts showed a significant pore size distribution graph in the range of less than 3 nm. It indicates the porous silica layer coated on the Ni/SiO₂ catalyst surface. During additional outer SiO₂ shell coating step, the surfactant (CTAB) was used as pore directing agents. The dissolved CTAB molecules form the micelles in aqueous suspension and the sequential hydrolysis-condensation reaction of TEOS occurs outside of the micelle during coating of outer SiO₂ shell on pre-synthesized Ni/SiO₂ catalysts. After the subsequent calcination step, residual pore directing agent (CTAB) is removed and pores are generated in the outer SiO₂ shell layer. Both TEM images (Fig. 1) and N₂ physisorption results (Fig. 2) confirmed that core-shell nanostructures are well formed by coating porous silica layer on the Ni/SiO₂ catalyst surface.



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Fig. 2. (a) N₂ physisorption isotherm and (b) pore size distribution of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts.

Table 1. Physical properties of catalysts.

Sample	Ni loading [wt %] [a]	Surface area [m ² /g] [b]	Pore volume [cm ³ /g] [b]	Pore size [nm] [b]	Ni crystalline size [nm] [c]	Ni dispersion [%] [d]
Ni/SiO ₂ _WI	5.1	13.4	0.05	35.2	8.7	0.40
Ni/SiO ₂ _AE	3.6	40.4	0.08	11.8	3.9	4.87
Ni/SiO ₂ _WI@SiO ₂	3.9	491	0.09	2.3	11	0.25
Ni/SiO ₂ _AE@SiO ₂	2.5	421	0.12	2.5	3.1	2.95

[a]

Measured by ICP-OES.

[b]

Measured using N₂ adsorption–desorption.

[c]

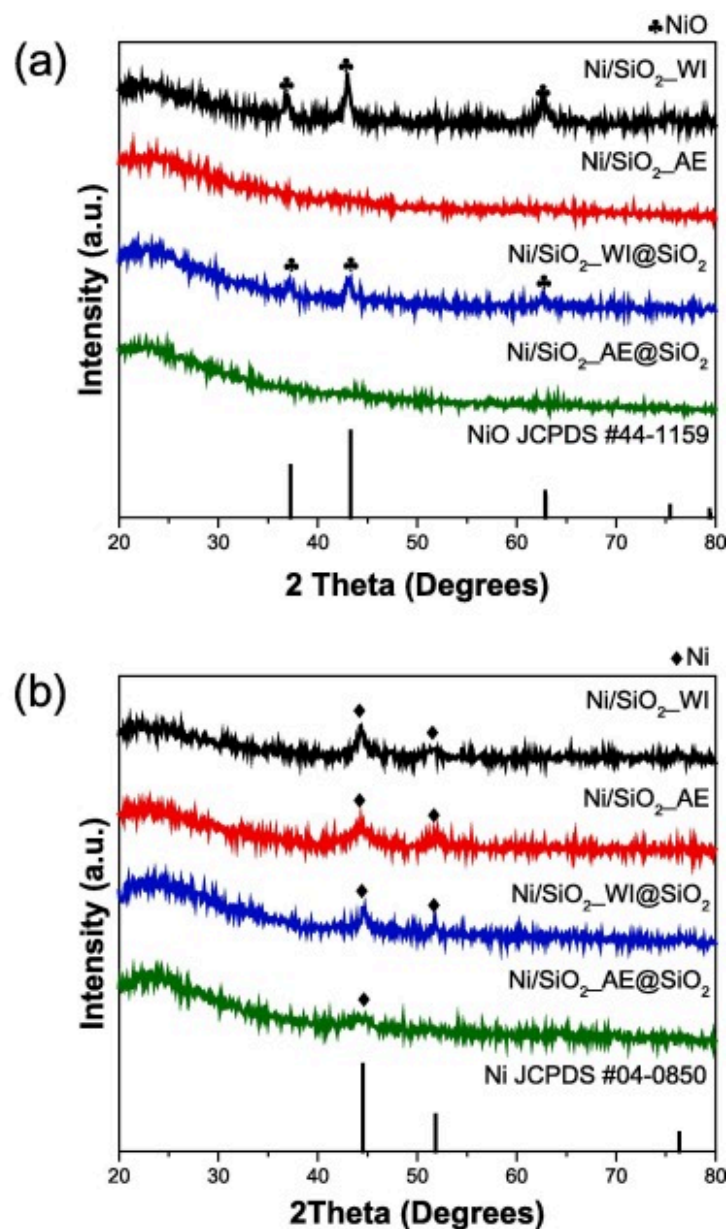
Calculated from XRD by scherrer's equation of Ni (111).

[d]

Measured by H₂ pulse chemisorption

Fig. 3 shows the XRD patterns of the calcined catalysts and the reduced catalysts, respectively. As shown in Fig. 3(a), all samples show a broad peak at $2\theta=22^\circ$ indicating the amorphous silica. After wet impregnation followed by calcination of the Ni/SiO₂_WI catalyst, significant peaks corresponding to the (111), (200), and (220) planes of NiO are shown at $2\theta=37.2, 43.3,$ and 62.9° . On the other hand, Ni/SiO₂_AE synthesized by ammonia evaporation did not show any obvious peak. It is well known that the impregnated nickel species can be easily converted to NiO particles and sintered simultaneously on the surface of silica during calcination (Yoo et al., 2023). However, since nickel phyllosilicate can be formed with low crystallinity during ammonia

evaporation followed by calcination, Ni/SiO₂_AE does not exhibit any obvious crystalline peak. In the case of the Ni/SiO₂_WI@SiO₂ core-shell catalyst, the peak intensity of NiO significantly decreased due to the presence of additionally introduced silica layer. The Ni/SiO₂_AE@SiO₂ core-shell catalyst did not exhibit a distinct peak like the Ni/SiO₂_AE catalyst.



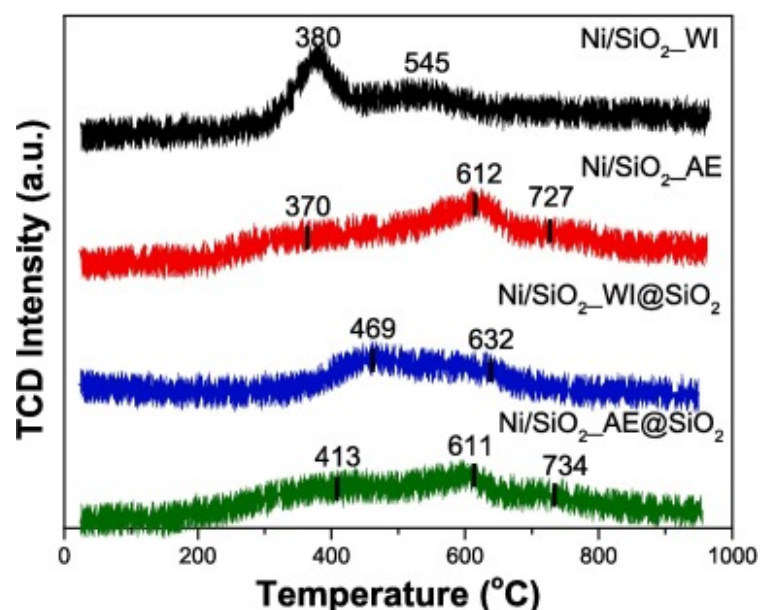
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Fig. 3. XRD pattern of (a) freshly calcined catalysts and (b) reduced catalysts.

After reduction of the prepared Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts (Fig. 3(b)), NiO of the Ni/SiO₂_WI and Ni/SiO₂_WI@SiO₂ catalysts was reduced and the peaks corresponding to the (1 1 1) and (200) planes of metallic nickel were observed at $2\theta=44.5, 51.8^\circ$, respectively. In the Ni/SiO₂_AE and Ni/SiO₂_AE@SiO₂ catalysts, nickel phyllosilicate was also reduced to form metallic nickel. As shown in Fig. 3(b), the intensity and sharpness of metallic nickel over Ni/SiO₂_WI and Ni/SiO₂_WI@SiO₂ are stronger and sharper than those over Ni/SiO₂_AE and Ni/SiO₂_AE@SiO₂, respectively. The nickel crystal sizes of Ni/SiO₂_WI and Ni/SiO₂_WI@SiO₂ catalysts prepared by wet impregnation were 8.7 and 11 nm, respectively, which are larger than that of Ni/SiO₂_AE and Ni/SiO₂_AE@SiO₂ catalysts (3.9 and 3.1 nm, respectively). The overall nickel crystal size of all catalysts calculated by Scherrer equation was smaller than the nickel particle size characterized by TEM (Fig. 1). Calculation by Scherrer equation from XRD peak gives us average value of Ni crystal size, not individual particle size. Additionally, the crystal size calculated from XRD represents the overall average size of metal crystal, while TEM images show particle sizes of metal particles which are very locally distributed. The calculated values of nickel loading amount of Ni/SiO₂_WI, Ni/SiO₂_AE, Ni/SiO₂_WI@SiO₂, and Ni/SiO₂_AE@SiO₂ were 5.1, 3.6, 3.9, and 2.5wt%, respectively (Table 1). Due to additional silica coating, nickel loadings of core-shell nanostructured catalysts were lower than those of their other catalysts. The nickel dispersion of Ni/SiO₂_WI, Ni/SiO₂_AE, Ni/SiO₂_WI@SiO₂, and Ni/SiO₂_AE@SiO₂ catalysts calculated by H₂ chemisorption was 0.40, 4.87, 0.25, and 2.95%, respectively (Table 1). Consistent with TEM (Fig. 1) and XRD (Fig. 2(b)) analysis, the dispersion of Ni/SiO₂_AE was higher than that of Ni/SiO₂_WI. On the other hand, the core-shell catalysts have lower dispersion than their parent catalysts. In Table S1, it shows H₂ adsorption normalized by the weight of catalysts and the weight of Ni. After SiO₂ shell coating process, H₂ adsorption quantity is significantly reduced in both Ni/SiO₂_WI and Ni/SiO₂_AE catalysts. It is due to the outer silica shell covers the surface of nickel nanoparticles despite of the considerable the porosity of the outer silica shell. As a result, outer silica shell partially blocks the probe molecules (H₂) to access the metal surface.

The H₂-TPR profile of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts are shown in Fig. 4. As the different synthetic methods are applied, the H₂-TPR profiles of Ni/SiO₂_WI and Ni/SiO₂_AE are different. Ni/SiO₂_WI has two reduction peaks with a main peak near 380 °C. The reduction peak in the range of 300 to 450 °C corresponds to the reduction of NiO species that interacts weakly with silica, and the broad shoulder in the range of 450 to 600 °C are responsible to the reduction of NiO that interacts moderately with the support (Mihet et al., 2021). On the other hand, Ni/SiO₂_AE has three reduction peaks and the main peak was shown around

612 °C. The peak above 600 °C corresponds to the reduction of nickel phyllosilicate, which strongly interacts with the silica support (Zhang et al., 2015). Based on the H₂-TPR of Ni/SiO₂, Ni species in Ni/SiO₂_WI mainly exists in the form of NiO, but Ni species in Ni/SiO₂_AE are in the form of nickel phyllosilicate, which strongly interacts with the silica support. In addition, Ni/SiO₂_AE has a higher overall reduction temperature than Ni/SiO₂_WI, indicating that the ammonia evaporation can enhance the metal-support interaction of the resulted supported metal catalyst than wet impregnation. Previously, M. Yang et al. (Yang et al., 2015) reported that strong metal-support interaction can lower the reducibility of the metal oxide to form smaller metal particles. The low reducibility of nickel phyllosilicate in this study made it possible for Ni/SiO₂_AE to form small nickel nanoparticles. It is consistent with the TEM and XRD results shown in Fig. 1, Fig. 2. Ni/SiO₂_WI@SiO₂ core-shell catalyst exhibits similar H₂-TPR profile compared to that of the parent catalyst, but the reduction peaks were slightly shifted to a higher temperature. Q.Zhang et al. (Zhang et al., 2015) reported that the nickel species inside the channel of SBA-15 showed stronger metal-support interactions than those outside the channel due to the confinement effect. In our study, the reduction degree of nickel species is decreased due to the confinement effect reported in previous references. Ni/SiO₂_AE@SiO₂ core-shell catalyst also showed the slightly shifted peaks toward high temperature in H₂-TPR profiles compared to Ni/SiO₂_AE catalyst due to similar confinement effect.



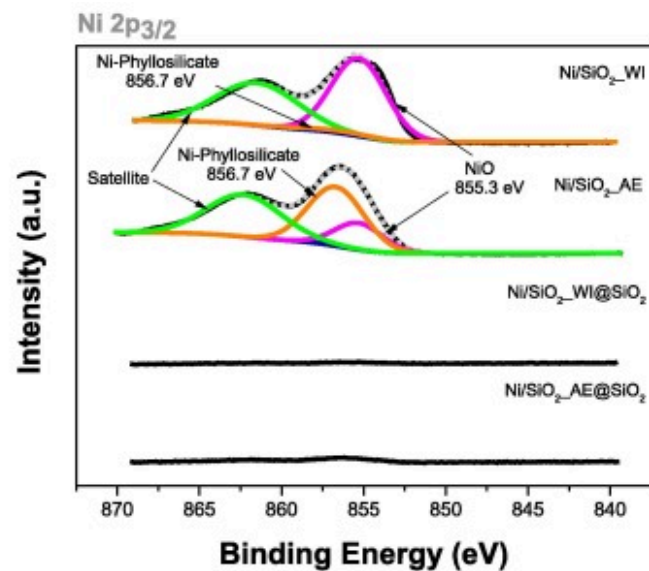
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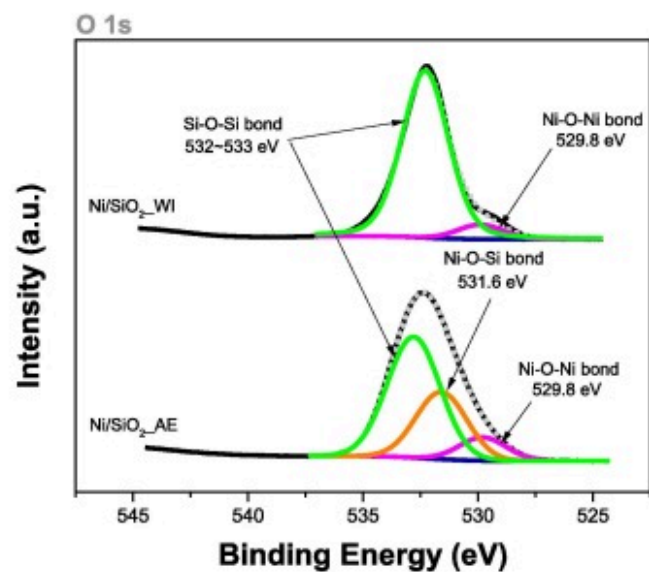
Fig. 4. H₂-TPR profile of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts.

Fig. 5 shows the results of XPS analysis of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts after calcination and reduction. In the Ni 2p XPS spectra (Fig. 5(a)) of the catalyst after calcination, Ni/SiO₂_WI and Ni/SiO₂_AE showed the peaks related to NiO (Lu et al., 2021) and nickel phyllosilicate (Han et al., 2020) at around 855.3 and 856.7 eV, respectively. On the other hand, Ni/SiO₂_WI@SiO₂ and Ni/SiO₂_AE@SiO₂ core-shell catalysts did not show any obvious peak corresponding to nickel. This should be due to the presence of the outer silica shell with a thickness of about 20 nm confirmed by the TEM (Fig. 1) result. In the TEM result, the outer silica shell is quite thicker than the normal XPS detection depth of 10 nm (Ratner and Castner, 2009). Fig. 5(b) shows the O 1s XPS spectra of the catalyst after calcination. Ni/SiO₂_WI has a strong peak related to Si-O-Si bonds due to silica nano spheres and a minor peak related to Ni-O-Ni bonds of NiO. Whereas Ni/SiO₂_AE exhibited obvious three peaks, which corresponded to the Si-O-Si, Ni-O-Ni and Ni-O-Si bonds, representing nickel phyllosilicate. From the results of the XPS analysis of the calcined catalysts, Ni species of Ni/SiO₂_WI exists in the form of NiO, and that of Ni/SiO₂_AE mainly exists in the form of nickel phyllosilicate with major portion. It is well known that the Ni-O-Si bonds from nickel phyllosilicate allow stronger metal support interactions than Ni-O-Ni bonds from NiO. So, it can be easily expected that Ni/SiO₂_AE has stronger metal-support interactions characteristics than Ni/SiO₂_WI, which is consistent with the results of the H₂-TPR analysis shown in Fig. 4. In case of Ni/SiO₂_AE catalyst, the Ni-O-Si bond that exists at the interface between the nickel species and silica support acts as an anchor, preventing metal particles from sintering and maintaining a high dispersion state, which is confirmed in both TEM (Fig. 1) and XRD (Fig. 3) results. On the other hand, in case of Ni/SiO₂_WI, the interface between nickel species and support cannot prevent sintering of metal particles and does induce low dispersion state due to weak metal-support interaction. In the Ni 2p XPS spectra of the reduced Ni/SiO₂_WI and Ni/SiO₂_AE (Fig. 5(c)), a peak corresponding to Ni⁰ was observed at 853.0 eV, and peaks related to NiO and nickel phyllosilicate remained. The Ni⁰/(Ni⁰+Ni²⁺) ratio of the reduced Ni/SiO₂_WI and Ni/SiO₂_AE were 0.291 and 0.265, respectively, indicating that the Ni species of Ni/SiO₂_WI was more reduced. This is because the nickel phyllosilicate species with strong metal-support interaction and are difficult to be reduced, are mainly present in Ni/SiO₂_AE.

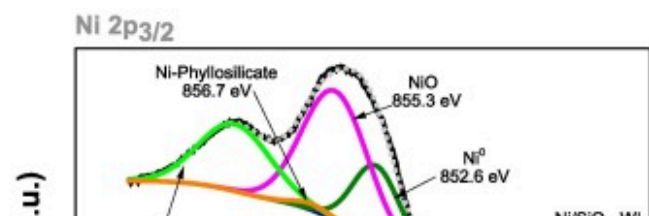
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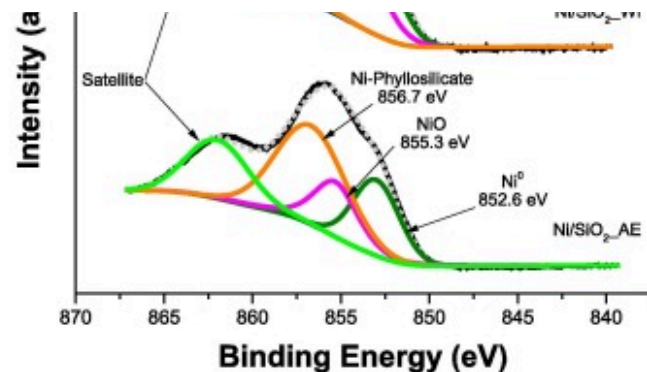


(b)



(c)





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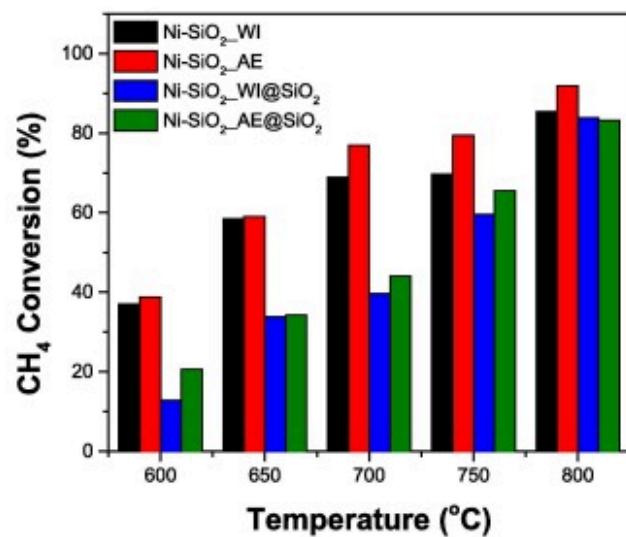
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Fig. 5. (a) Ni 2p, (b) O 1s XPS spectra of the calcined catalysts and (c) Ni 2p XPS spectra of reduced Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts.

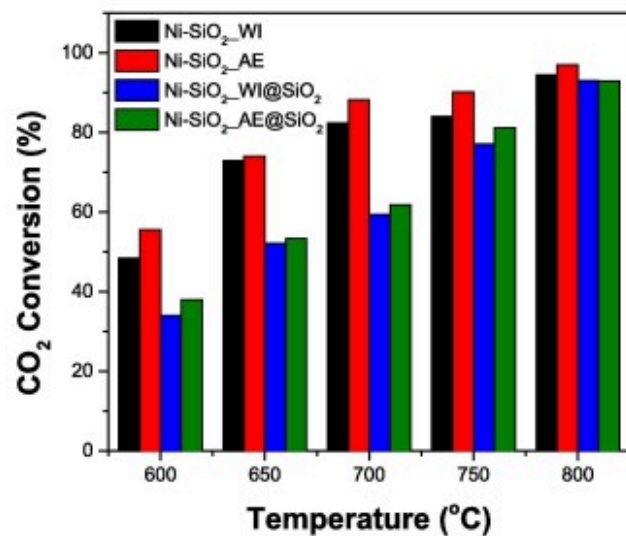
Catalytic performance of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalyst: The results of the catalytic activity test of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts in methane dry reforming are shown in Fig. 6. The methane dry reforming experiment was carried out with equimolar CH₄ and CO₂ diluted with an inert gas with 48000 ml/g^{*}h of WHSV. Since the methane dry reforming reaction is endothermic, the activity of all catalysts increased as the reaction temperature increased. Also, the CO₂ conversion is slightly higher than that of CH₄, indicating that a reverse water-gas shift reaction has slightly occurred (Cheng et al., 2006, Tomiyama et al., 2003). The activity of Ni/SiO₂-AE is higher than that of Ni/SiO₂-WI under all temperature conditions, and the syngas selectivity is also higher. This is due to the formation of highly dispersed nickel nanoparticles by the strong metal-support interaction shown in the previous H₂-TPR (Fig. 4) and XPS (Fig. 5) results. We also compared the specific activity which can tell us

the catalytic activity based on the normalized amount of Ni content on Ni/SiO₂_WI and Ni/SiO₂_AE based catalysts ([Fig. S3, supplementary information](#)). Although the Ni content of the Ni/SiO₂_AE based catalysts were lower than that of the Ni/SiO₂_WI based catalysts, the Ni/SiO₂_AE based catalysts showed higher specific activity than Ni/SiO₂_WI based catalysts due to their high Ni dispersion. Unlike the Ni/SiO₂ catalysts, Ni/SiO₂@SiO₂ core-shell catalysts exhibit much lower catalytic activity because the additionally introduced silica shell covers the active site and reduces the nickel content of the catalysts ([Yang et al., 2020](#)). This observation is consistent with the results of H₂-chemisorption and ICP-OES ([Table 1](#)). In addition, the mass transfer limitation by the porous silica shell is also a significant factor that adversely affects the catalytic activity. As can be seen from the result of [Fig. 2](#) and [Table 1](#), the additionally introduced silica shell has small mesopores with a diameter of about 2 nm. In the range of mesopores and micropores below 10 nm, Knudsen diffusion dominates ([Nonoyama et al., 2011](#)), and it is well known that a small pore size less than 2.5 nm is sufficient to induce diffusion limitation ([Bian and Kawi, 2018](#)).

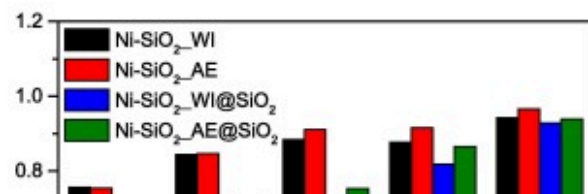
(a)

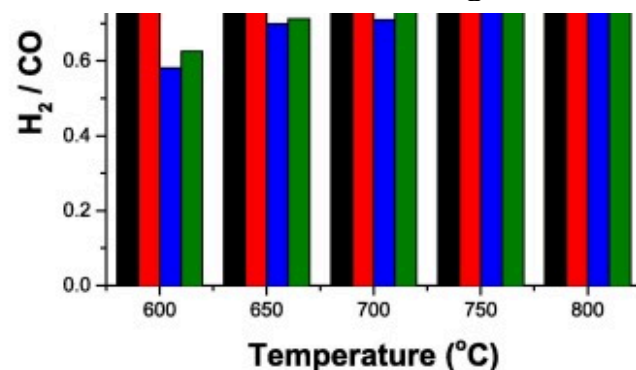


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(c)





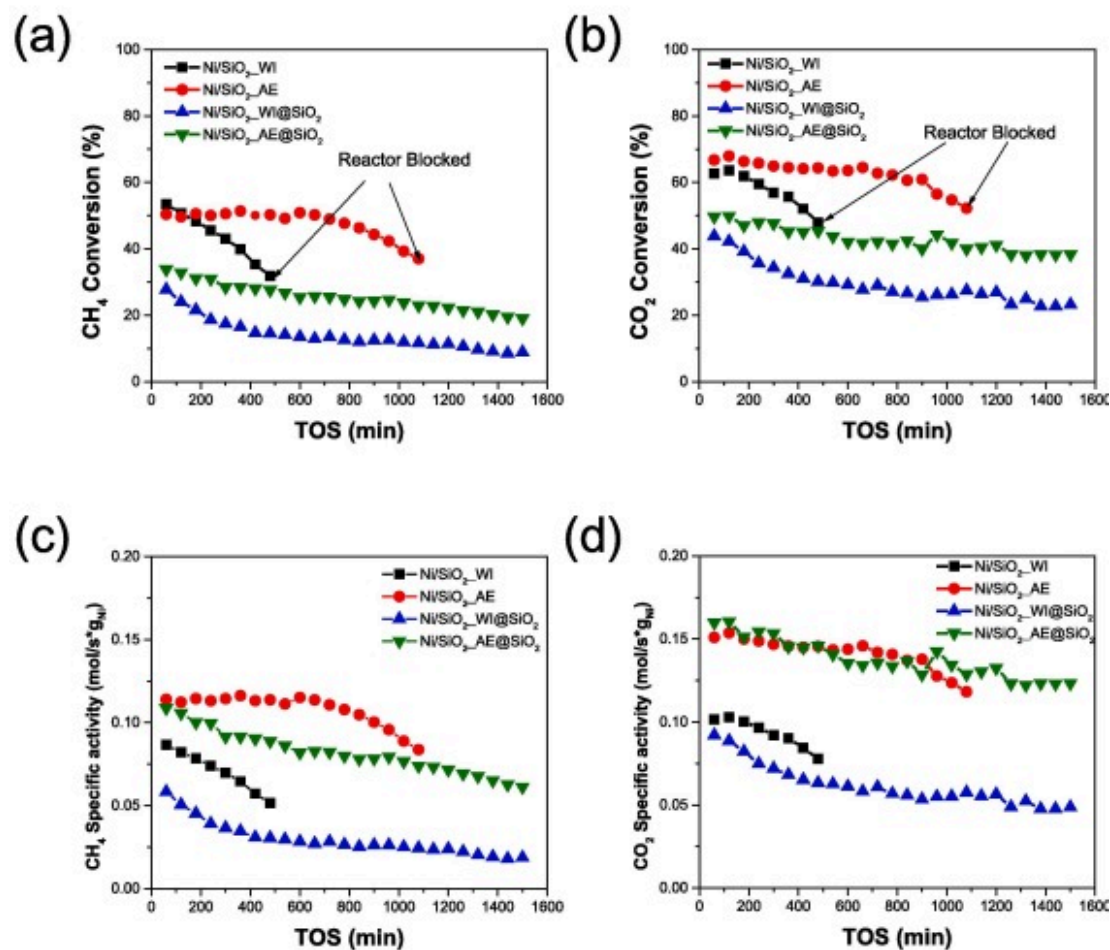
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Fig. 6. Catalytic activity of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts on dry reforming; (a) CH₄ conversion, (b) CO₂ conversion, (c) H₂/CO ratio (Reaction condition: reaction temperature 600°C~800°C, WHSV 48,000ml/h*gcatal).

According to the literature by C. Jensen et al. (Jensen and Duyar, 2021), coke formation can be promoted at a low temperature condition of methane dry reforming by thermodynamic equilibrium, resulting in catalyst deactivation. Therefore, to verify the enhanced coke resistance of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts, stability tests were performed at 600°C, and the results are shown in Fig. 7. Methane dry reforming was performed at atmospheric pressure and the reactor operation was shut down when the pressure at the front of the catalyst bed reaches 1 barg. At 600°C, Ni/SiO₂_WI showed rapid deactivation and the experiment was stopped at 500min after the start (Fig. 7(a and b)). Ni/SiO₂_AE maintained stable activity for 600min, but then gently deactivated and the experiment was stopped in 1100min. On the other hand, the Ni/SiO₂@SiO₂ core-shell catalysts had lower activity than the Ni/SiO₂ catalysts, but showed stable catalytic activity for more than 1500min. As shown in Fig. 7(c and d), the

specific activity of Ni/SiO₂_AE based catalysts was higher than that of the Ni/SiO₂_WI based catalysts. Higher specific activity of Ni/SiO₂_AE based catalysts was attributed to the highly dispersed nickel nanoparticles by strong metal support interaction. In [Fig. 7](#), the main reason for catalytic deactivation was supposed to be carbon deposition on nickel nanoparticles. It is known that carbon deposition on metal nanoparticles easily occurs by methane decomposition (Eq. (2)) and boudard reaction (Eq. (3)). And it is well known that carbon deposition is promoted by large metal particles ([Kim et al., 2000](#)). However, even catalysts with small nickel nanoparticles such as Ni/SiO₂_AE cannot avoid severe coking under harsh conditions as shown in [Fig. 7](#). The Ni/SiO₂_AE@SiO₂ catalyst shows small nickel phyllosilicate species, which are intrinsically protected by the outer silica shell. Thus, the nickel species can avoid thermal sintering and can be well maintained in the original particle size. The outer shell controls the reactant molecular diffusion resulting in negligible formation of carbon coke on the Ni species. This suggests that the additionally introduced silica shell had a very significant effect on suppressing coking.



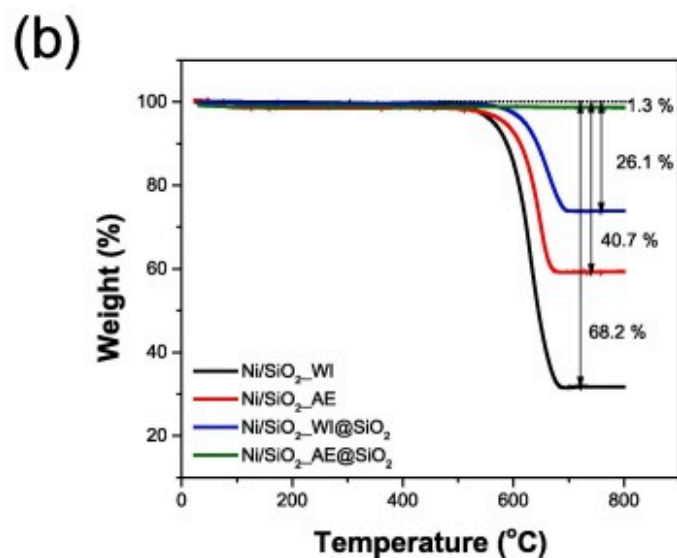
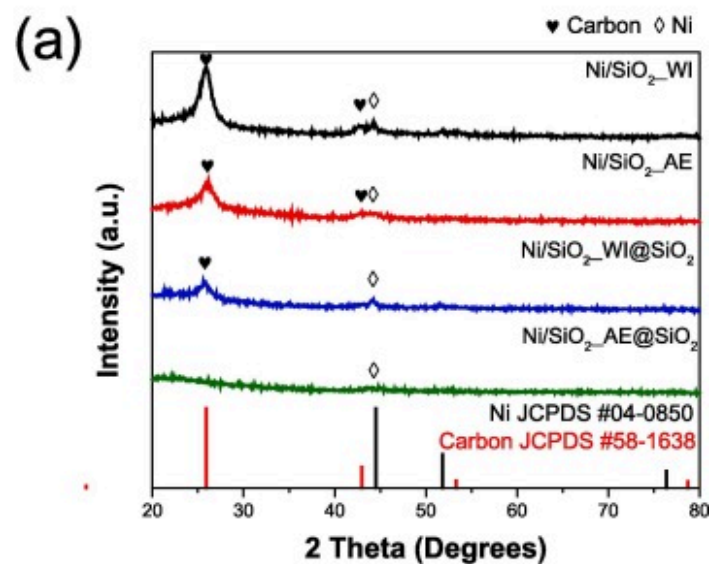
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Fig. 7. Catalytic stability of Ni/SiO₂ and Ni/SiO₂@SiO₂ catalysts on dry reforming; Conversion of (a) CH₄ and (b) CO₂, specific activity of (c) CH₄ and (d) CO₂ (Reaction Condition: reaction temperature 600 °C, WHSV 48,000 ml/h*g_{cat}).

Characterization of spent Ni/SiO₂ and Ni/SiO₂@SiO₂ catalyst: To investigate both the causes of catalyst deactivation and its degree, the spent catalysts were characterized by TGA, XRD, and TEM. The TGA results of spent Ni/SiO₂ and Ni/SiO₂@SiO₂ are

shown in Fig. 8(a). Ni/SiO₂_WI, Ni/SiO₂_AE, Ni/SiO₂_WI@SiO₂, and Ni/SiO₂_AE@SiO₂ showed weight reductions of ca. 68.2, 40.7, 26.1, and 1.3%, respectively, in the range of 500 to 700°C. Weight loss at high temperatures above 500°C indicates the presence of mostly filamentous carbon and graphitic carbon (Cao et al., 2021, Zhou et al., 2015). Interestingly, the carbon deposition amount of the Ni/SiO₂_WI@SiO₂ catalyst was very small compared to that of the Ni/SiO₂ catalyst, and Ni/SiO₂_AE@SiO₂ showed almost no weight loss. It means that additional porous silica shell can efficiently suppress the formation of coke during DRM reaction, and it allows the reactor operation to be durable for elongated period. In addition, when Ni particles are highly dispersed on the support surface and the metal-support interaction is considerably strong, the beneficial effects of outer porous shell are more obvious. As shown in Fig. 7, Ni/SiO₂_AE@SiO₂ exhibited even more durable characteristics with higher catalytic activity than Ni/SiO₂_WI@SiO₂. In addition, Ni/SiO₂_AE@SiO₂ showed negligible coke formation in the TGA results, indicating no obvious deactivation. In Fig. S4(a and c) and Table S2, TGA results of spent catalysts were shown after time on stream test at either 700°C and 800°C. As the reaction temperature increased, coke formation dramatically decreased for all catalysts. In dry reforming of methane reaction, it is well known that coke formation reaction is thermodynamically unfavorable in high temperature. In additions, as consistent with the result of 600°C, introducing a core-shell nanostructure and maintaining high Ni dispersion by strong metal-support interaction are important in suppressing coke formation. As shown in Fig. 8(b), peaks corresponding to crystalline carbon of MWCNTs formed during the reaction appear at $2\theta=26, 42^\circ$ for Ni/SiO₂_WI, Ni/SiO₂_AE, and Ni/SiO₂_WI@SiO₂ respectively. However, no peak was observed for Ni/SiO₂_AE@SiO₂, indicating negligible carbon species in the sample, which is consistent with the TGA results. After the dry reforming reaction at 700°C and 800°C, the peaks corresponding to coke are not observed (Fig. S4(b and d)). This is because the amount of deposited coke is less than the XRD detection limit and it is consistent with the TGA results (Fig. S4(a and c)).

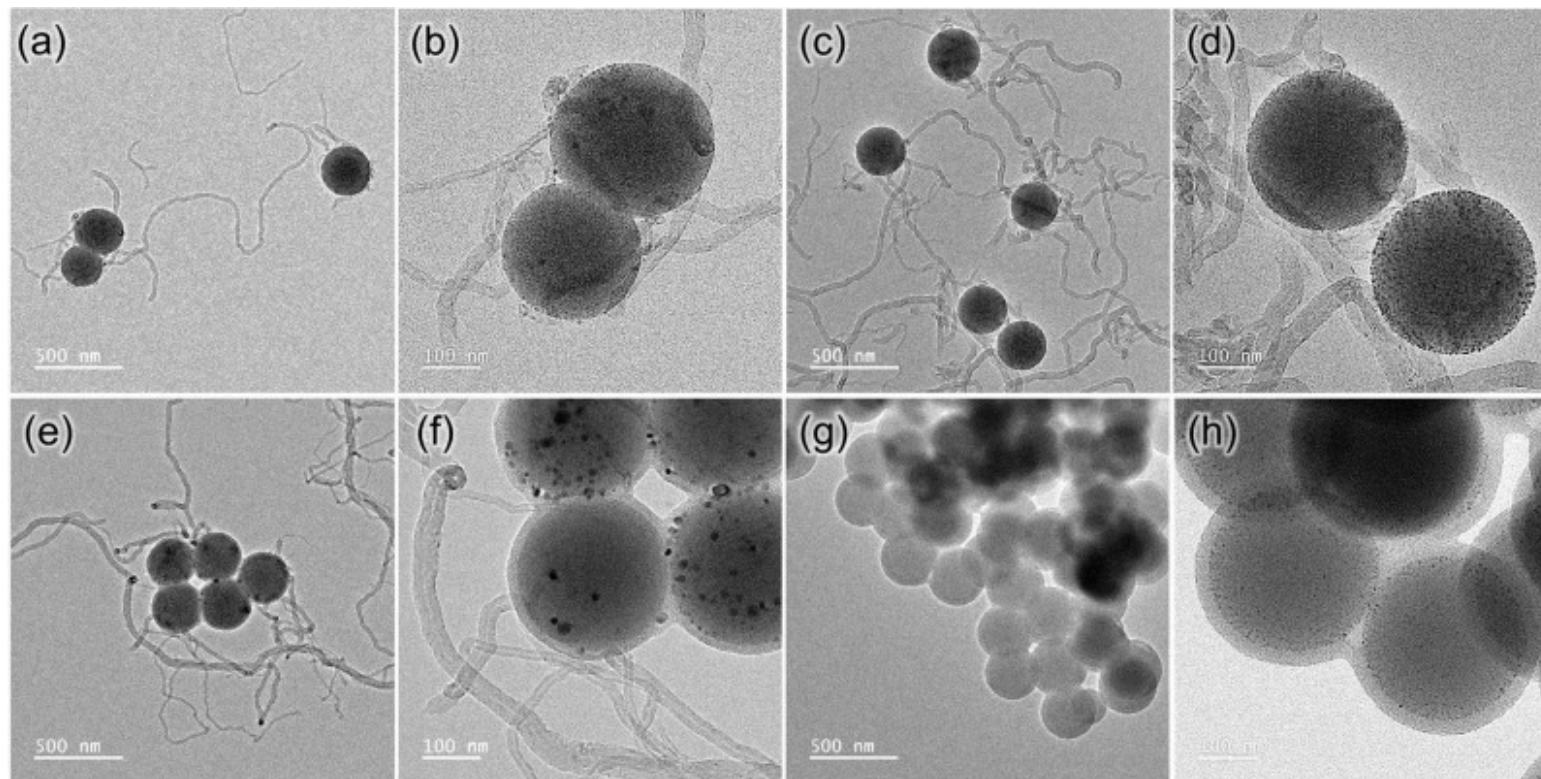


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Fig. 8. (a) XRD patterns and (b) TGA profiles of spent catalysts after 24h of TOS at 600°C.

To investigate the morphology of coke formation, we carried out TEM analysis for spent catalysts (Fig. 9). As shown in Fig. 9, a large amount of carbon nanotubes was formed in the Ni/SiO₂_WI, Ni/SiO₂_AE, and Ni/SiO₂_WI@SiO₂ catalysts. The formation of filamentous carbon does not directly deactivate the surface of the nickel nanoparticles, but causes pore plugging and catalyst bed expansion, which eventually completely blocks the reactor (Ochoa et al., 2020). On the other hand, no obvious carbon nanotubes were observed in the TEM image of Ni/SiO₂_AE@SiO₂ (Fig. 9(g and h)). This is consistent with the previous TGA and XRD analysis results (Fig. 8). According to the three characterization results of the spent catalysts, it can be confirmed that the durability against coke formation increases in the order of Ni/SiO₂_WI < Ni/SiO₂_AE < Ni/SiO₂_WI@SiO₂ < Ni/SiO₂_AE@SiO₂. Ni/SiO₂_WI and Ni/SiO₂_WI@SiO₂ catalysts induced more coke than Ni/SiO₂_AE and Ni/SiO₂_AE@SiO₂. In addition, the Ni/SiO₂@SiO₂ core-shell catalysts effectively suppressed coke formation than their parent catalysts. Nickel phyllosilicate, the main Ni species of Ni/SiO₂_AE based catalysts, has a stronger metal-support interaction compared to NiO, which is the main Ni species for Ni/SiO₂_WI based catalysts. On the small Ni nanoparticles, coke formation was effectively inhibited because methane decomposition reaction and boudard reaction, which are coke formation reactions, were less preferred. Also, unlike the Ni nanoparticles of the Ni/SiO₂ catalyst, Ni nanoparticles of Ni/SiO₂@SiO₂ significantly reduced coke formation due to the confinement effect in interacting with the porous silica shell as well as the silica spheres.



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Fig. 9. TEM image of (a, b) Ni/SiO₂_WI; (c, d) Ni/SiO₂_AE; (e, f) Ni/SiO₂_WI@SiO₂; (g, h) Ni/SiO₂_AE@SiO₂; after 24h of TOS at 600°C.

4. Conclusions

The Ni/SiO₂ catalyst was prepared by wet impregnation and ammonia evaporation. Ni/SiO₂_AE showed stronger metal-support interaction than Ni/SiO₂_WI through formation of nickel phyllosilicate, which was confirmed by the H₂-TPR and XPS analysis. The formation of nickel phyllosilicate induces a strong interaction between Ni²⁺ and SiO₂, which reduces the reducibility of the deposited nickel and consequently allows the formation of highly dispersed metallic nickel nanoparticles after reduction. The Ni/SiO₂@SiO₂ core-shell catalyst was successfully synthesized by coating a porous silica layer over the pre-synthesized

Ni/SiO₂_WI and Ni/SiO₂_AE. The introduced porous silica shell enhanced the metal-support interaction through additional interaction with the inner nickel species. From the result of the catalytic activity test, it is observed that the catalyst synthesized based on ammonia evaporation showed higher catalytic activity than the wet impregnation-based catalyst due to the highly dispersed active metallic nickel sites. On the other hand, the Ni/SiO₂@SiO₂ core-shell catalyst showed low catalytic activity because of the mass transfer limitation by the additional porous silica shell and the encapsulated active metallic nickel site between the silica core and porous shell. In the catalyst durability experiments, the Ni/SiO₂_AE@SiO₂ catalyst was the most stable and produced the least coke due to the strong metal-support interaction and the introduction of the silica shell.

CRedit authorship contribution statement

Dong Seop Choi: Writing – original draft, Investigation, Data curation, Conceptualization. **Na Yeon Kim:** Methodology, Investigation. **Eunju Yoo:** Methodology, Investigation, Formal analysis. **Jiyull Kim:** Validation, Methodology. **Ji Bong Joo:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ji Bong Joo reports financial support was provided by Konkuk University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

The following are the Supplementary data to this article:

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Supplementary Data 1.

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Data availability

No data was used for the research described in the article.

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